explain linear vs nonlinear difference equations

Linear difference equations are those where the unknown sequence and its shifts (like $x_{n+1}, x_n, x_{n-1}, \ldots$) appear only to the first power, are not multiplied together, and have coefficients that do not depend on the sequence itself. Nonlinear difference equations violate at least one of these conditions.

What "linear" means here

For a scalar sequence $\{x_n\}$, a k-th order linear difference equation has the form

$$a_k(n)x_{n+k} + a_{k-1}(n)x_{n+k-1} + \cdots + a_0(n)x_n = b(n),$$

where $a_i(n)$ and $b(n)$ are known functions of n only, and each x_{n+j} appears linearly (power 1, no products like $x_n x_{n-1}$). [1] [2]

Key properties:

- **Superposition** holds: if $x_n^{(1)}$ and $x_n^{(2)}$ solve the homogeneous equation ($b(n) = 0$), then any $c_1 x_n^{(1)} + c_2 x_n^{(2)}$ is also a solution. [2]
- Solutions can often be found systematically (e.g., characteristic equations for constant coefficients). [1] [2]

Examples (linear):

- First order: $x_{n+1} = 2x_n + 3$ (or $x_{n+1} - 2x_n = 3$).
- Second order: $x_{n+2} - 5x_{n+1} + 6x_n = 0$.
  In both, each term is a linear combination of x_{n+j} with coefficients depending at most on n.

What makes it nonlinear

A difference equation is nonlinear if the sequence terms appear with powers other than 1, in products, in denominators, or inside nonlinear functions. [2] [1]

Common nonlinear features:

- Powers: x_n^2, $\sqrt{x_n}$.
- Products: $x_n x_{n-1}$, $x_n^2 x_{n-1}$.
- Nonlinear functions: $\sin(x_n)$, $\exp(x_n)$, $\frac{1}{x_n}$.

Examples (nonlinear):

- Logistic map: $x_{n+1} = r x_n(1 - x_n)$ (product of x_n with itself and with $1 - x_n$).

- Quadratic term: $x_{n+1} = x_n^2 + 1$.
- Reciprocal: $x_{n+1} = \frac{1}{1+x_n}$.
  Each violates the linearity conditions by having products or nonlinear functions of x_n.

Why the distinction matters

- Linear equations are typically **easier to solve**, with well-developed closed-form and stability theory (e.g., via eigenvalues/characteristic roots).[1] [2]
- Nonlinear equations can show rich behavior (multiple equilibria, limit cycles, chaos) and **often lack closed-form solutions**, so one relies on qualitative and numerical analysis.[2] [1]

In short: check whether the unknown sequence terms appear only linearly with coefficients independent of those terms; if yes, the difference equation is linear, otherwise it is nonlinear.

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1. https://www.geeksforgeeks.org/maths/difference-between-linear-and-non-linear-equations/
2. https://math.libretexts.org/Bookshelves/Calculus/Differential_Calculus_for_the_Life_Sciences_(Edelstein-Keshet)/13:_Qualitative_Methods_for_Differential_Equations/13.01:_Linear_and_Nonlinear_Differential_Equations
3. https://study.com/academy/lesson/how-to-recognize-linear-functions-vs-non-linear-functions.html
4. https://www.youtube.com/watch?v=PMlWjkvmS9M
5. https://byjus.com/maths/difference-between-linear-and-nonlinear-equations/
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7. https://www.storyofmathematics.com/linear-vs-nonlinear-function/
8. https://www.thethinkacademy.com/blog/identifying-linear-vs-nonlinear-functions/
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